PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

AutiWise - A comprehensive digital toolkit for individuals with Autism Spectrum Disorder

Group No. 10

Arya Patil	22107047
Nisha Patel	22107037
Prachi Pawar	22107018
Renuka Udugade	22107048

Project Guide

Ms. Poonam Pangarkar
Ms. Shubhangi Soni

Contents

- Abstract
- Introduction
- Objectives
- Literature Review
- Research Gap
- Problem Definition
- Scope
- Technological Stack
- Proposed System Architecture/Working/Module Diagram
- Prototype Design Demonstration
- Implementation Status
- Review Suggestions (Given in Last meeting)

Abstract

AutiWise is an AI-powered digital system designed to provide early detection and continuous support for individuals with Autism Spectrum Disorder (ASD). It integrates four modules: a diagnostic tool for identifying ASD indicators, a conversational chatbot for communication and emotional support, a voice-command routine tracker to reduce anxiety and improve time management, and a social skills coach that generates AI-driven conversational scripts for role-play and learning. By combining AI with an intuitive interface, AutiWise aims to support individuals with ASD through timely intervention, improved communication, and enhanced social inclusion.

Sustainable Development Goals(SDG) mapped

- SDG 3: Good Health and Well-being - Promotes early detection and continuous support for individuals with autism, improving mental health and quality of life.
- SDG 4: Quality Education - Supports inclusive education through social skill-building and communication tools tailored for neurodiverse learners.
- SDG 10: Reduced Inequalities - Empowers individuals with ASD by providing accessible, personalized technology that supports independent living and social integration.

Introduction

- Autism Spectrum Disorder (ASD) is increasing worldwide, impacting communication, social interaction, and daily routines of affected individuals.
- This creates a strong need for an affordable, AI-powered, and interactive system that can deliver timely detection and continuous support for individuals with ASD.

Motivation

- Many individuals with ASD still struggle with expressing emotions, building social connections, and maintaining structured routines in daily life.
- Despite progress in therapies, there remains a gap for accessible, adaptive, and technology-driven solutions that provide continuous support.

Objectives

- To design and implement an AI-powered system for the early detection of Autism Spectrum Disorder (ASD).
- To develop a conversational chatbot that facilitates communication support and emotional expression for individuals with ASD.
- To build a voice-command-based routine tracker that helps users create and follow structured daily schedules to reduce anxiety and improve time management.
- To create a social skills coaching module that uses AI to generate real-world conversation scripts for role-play and social learning.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
1	Using Artificial Intelligence to Improve Empathetic Statements in Autistic Adolescents and Adults: A Randomized Clinical Trial [1]	Koegel, L.K., Ponder, E., Bruzzese, T. et al	2025	The paper uses a methodology of LLM-based AI hat prompts users and gives feedback on empathetic responses.	Focuses only on empathetic verbal responses; lacks integration with daily schedules.
2	Examining Virtual Reality Interventions for Social Skills in Children with Autism Spectrum Disorder: A Systematic Review [2]	Altın, Y., Boşnak, Ö. & Turhan, C	2025	Systematic review of VR social skills interventions in children with ASD.	VR hardwares are expensive. VR environments can cause autistic person more anxiety, disorientation, or cybersickness.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
3	AI-Powered Autism Detection, Awareness, and Social Integration: A Comprehensive Framework [3]	R. S. Ali Fathima, R. Gokkul, M. Nirmalsahaj, S. Rohith and S. Rupes	2025	This paper uses ruke-based chatbot to provide conversational support.	The chatbot is unable to answer outside its scope.
4	Towards an automatic early screening system for autism spectrum disorder in toddlers based on eye-tracking.[4]	Fernandez-Lanvin, D., Gonzalez-Rodriguez, M., De-Andres, J. et al.	2023	This research involves eye-tracking video analysis, feature extraction of gaze patterns, matched-pairs experimental design, ML classification	The drawback is that it needs specialized eye-tracking hardware and has limited robustness across environments.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
5	Computer Vision-based Interactive Autism Detection System using Deep Learning [5]	B. Parvej, SMM Alam, F. I. Fahim, MN Pathan, MA Rahaman	2024	Detection of autism traits on image data	Low accuracy, gives incorrect result in few scenario.
6	Comparative exploration of machine learning models for enhanced autism detection[6]	C. Nagamani, T. R. Devi, G. G. Babu, V. Sindhu	2024	Compares different ML models on datasets to detect autism	Likely based on questionnaire or structured data

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Gap Identified
7	Detection of Autism Spectrum Disorder in Children Using Machine Learning [7]	Vakadkar, K., Purkayastha, D. & Krishnan, D.	2021	This paper uses ML classifiers (SVM, Random Forest, Logistic Regression) applied to questionnaire datasets.	The drawback is that ML models use structured questionnaire data limiting real world applications.

Research Gap(Limitations of existing systems)

- Poor Integration with Therapists/Educators: Many toolkits function in isolation, without collaborative dashboards for therapists.
- Limited Detection: Most toolkits follow a one-size-fits-all approach. They fail to detect based on the child's individual needs, severity level, age.
- Narrow Focus of Functionality: Lack of integration of multiple supports (e.g., early detection + social coaching + emotional support).

Justification:

By leveraging a unified AI platform, we will create an accessible, and appropriate solution. This approach provides a comprehensive toolkit for early detection and continuous support through an integrated chatbot, routine tracker, and social skills coach.

Problem Definition

Individuals with Autism Spectrum Disorder (ASD) face significant challenges in early diagnosis, effective communication, and social interaction. Current digital tools are often fragmented, addressing only one aspect of support. This leads to an inconsistent user experience and fragmented care. AutiWise seeks to solve this by developing a single, AI-powered digital toolkit that provides a holistic solution for individuals with ASD and their caregivers.

Proposed Focus Areas-

- Image and Behavioral-based ASD Detection
- Conversational Chatbot for Assistive Communication
- Voice-Enabled Routine Planning and Reminders
- Social Script Generation for Role-Play and Social Skills Training
- Unified Platform for Long-Term ASD Support

Problem Definition

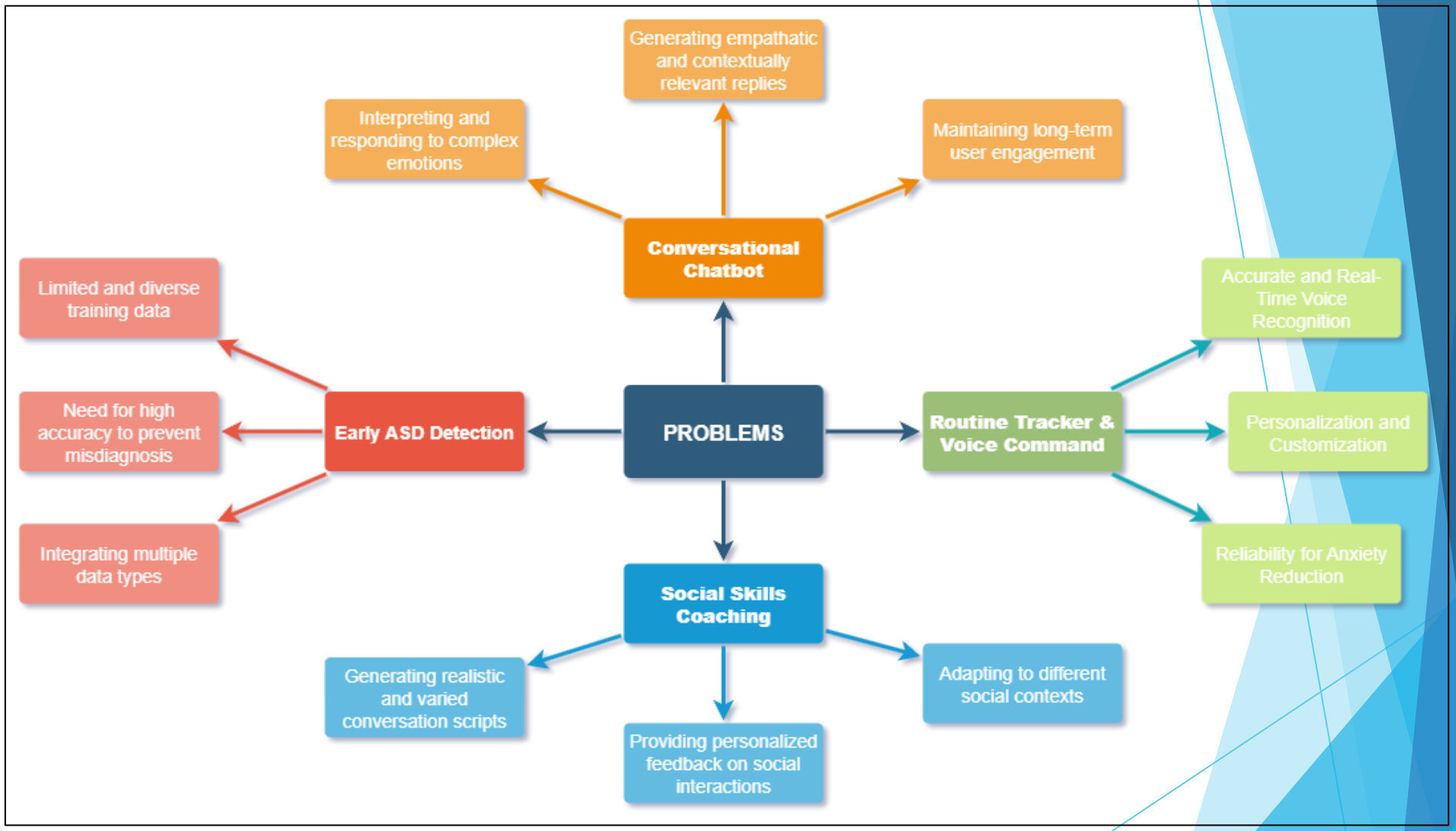


Figure 1: Problem Definition Diagram

Scope

- Develop an AI-powered system for both early detection and continuous support for individuals with Autism Spectrum Disorder (ASD).
- For ASD individuals , to provide constant support.
- An assistance platform for therapists to get the reports of ASD individuals and track routines.

Technological Stack

Front – end

- HTML5 / CSS3 / JavaScript: For building responsive and accessible user interfaces

Models

- OpenAI : For chatbot conversations and social script generation
- HaarCascade Classifier: To find precise location of a face in the image
- ResNet50: For ASD detection
- Speech Recognition (DeepSpeech / Vosk): For enabling voice-command functionality

Technological Stack

Datasets:

- Autistic Chidren Facial Dataset

Trained on ResNet50 model

Source- Kaggle

Analytics:

- Progress report for autistic individual

Proposed system architecture

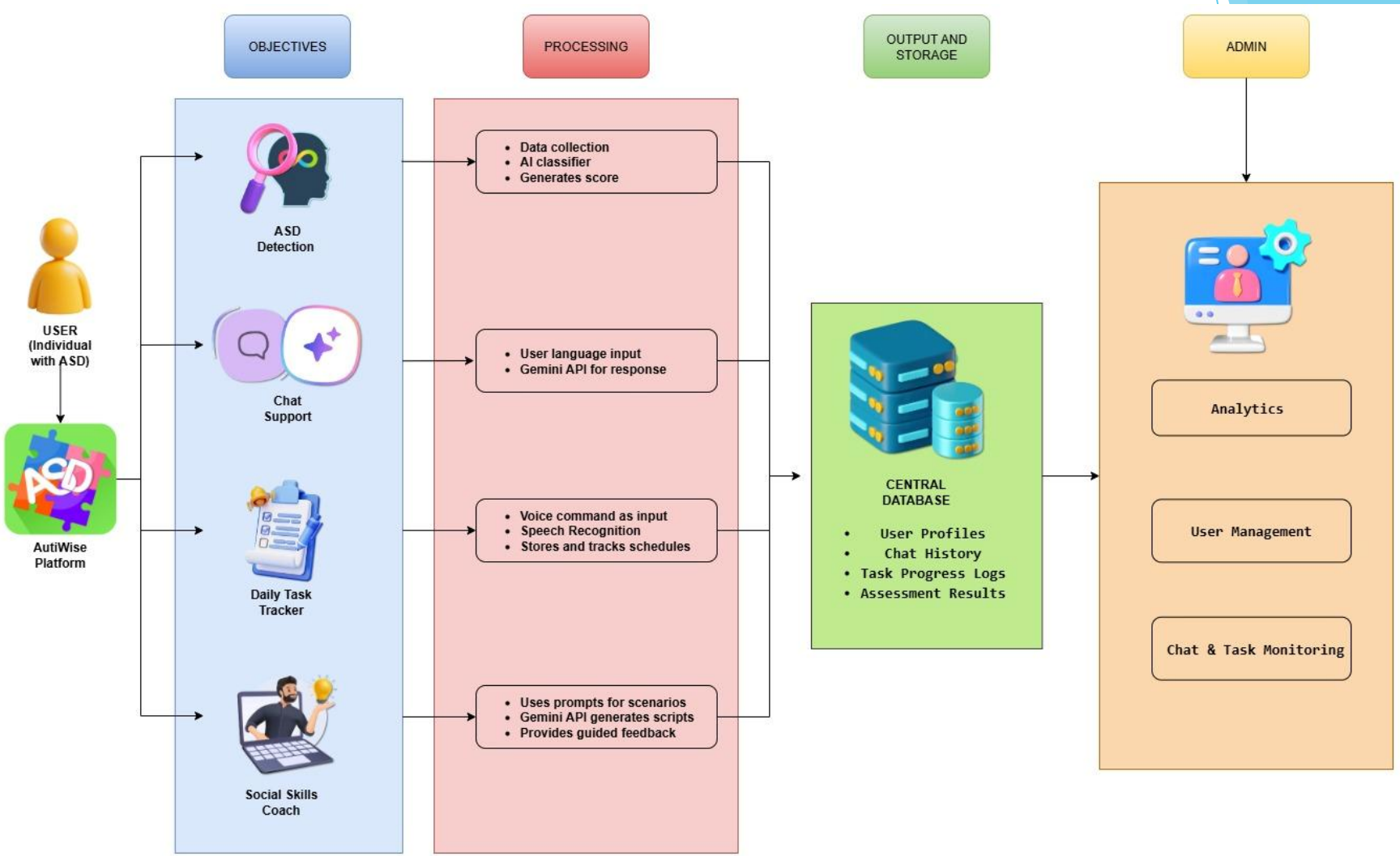


Figure 2: Proposed System Diagram

Prototype Design Demonstration

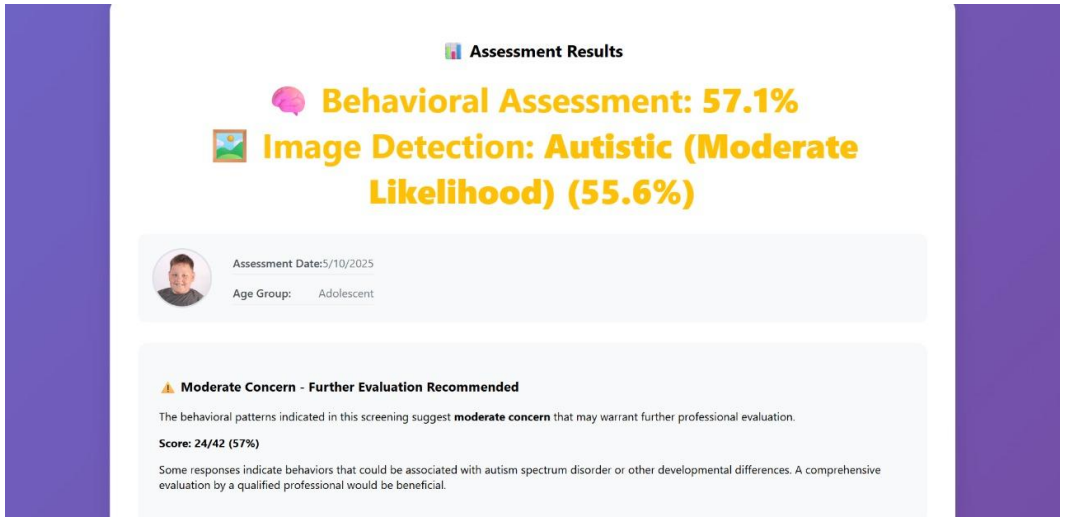


Figure 3: Detection result

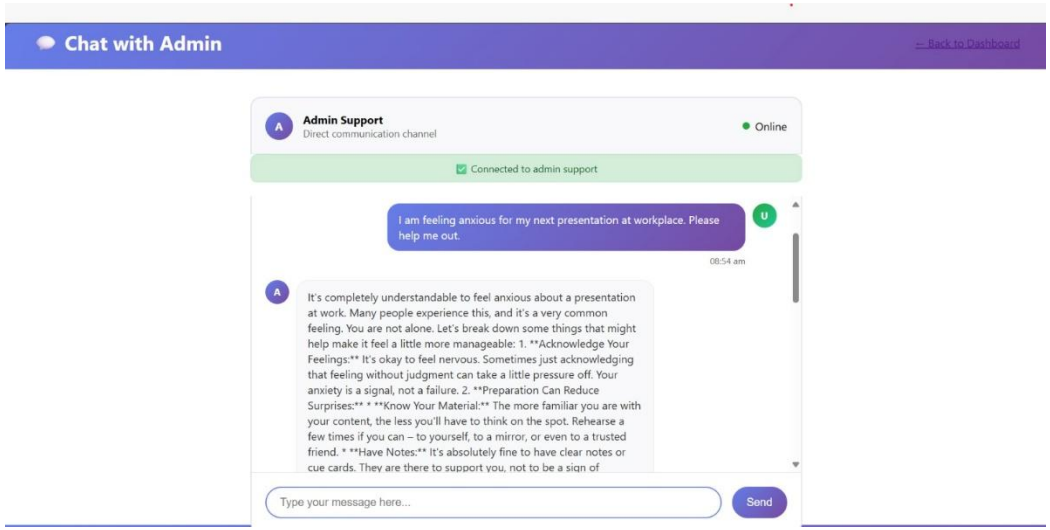
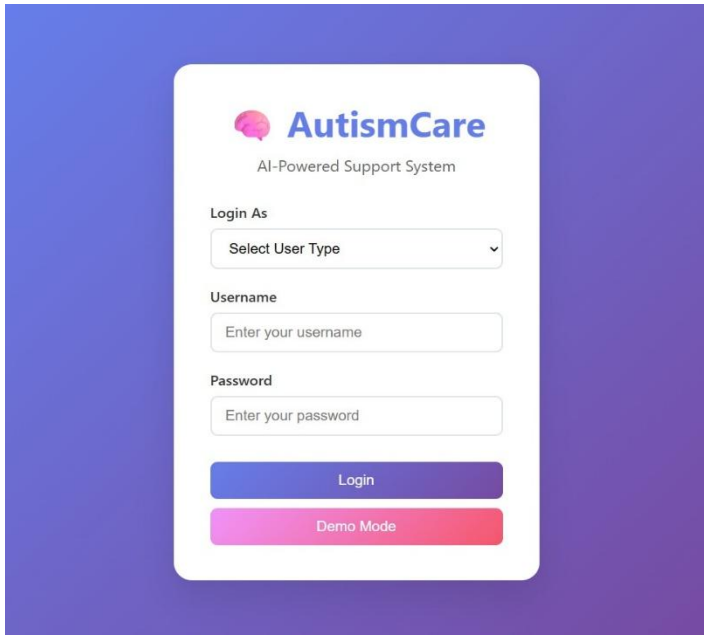
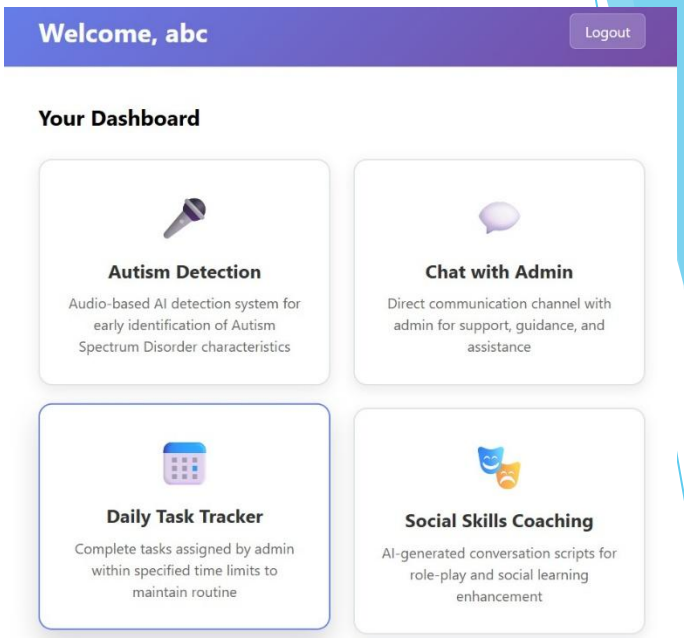
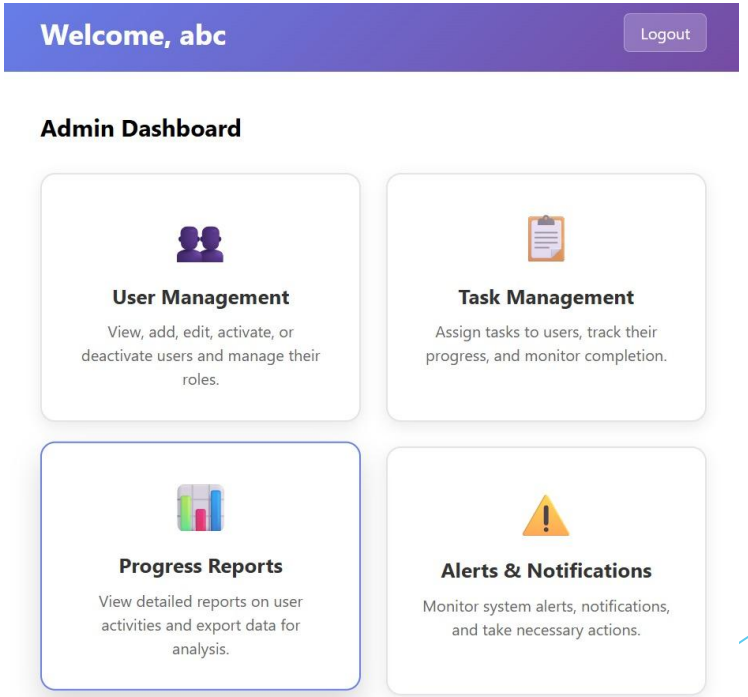


Figure 4: Chatbot result

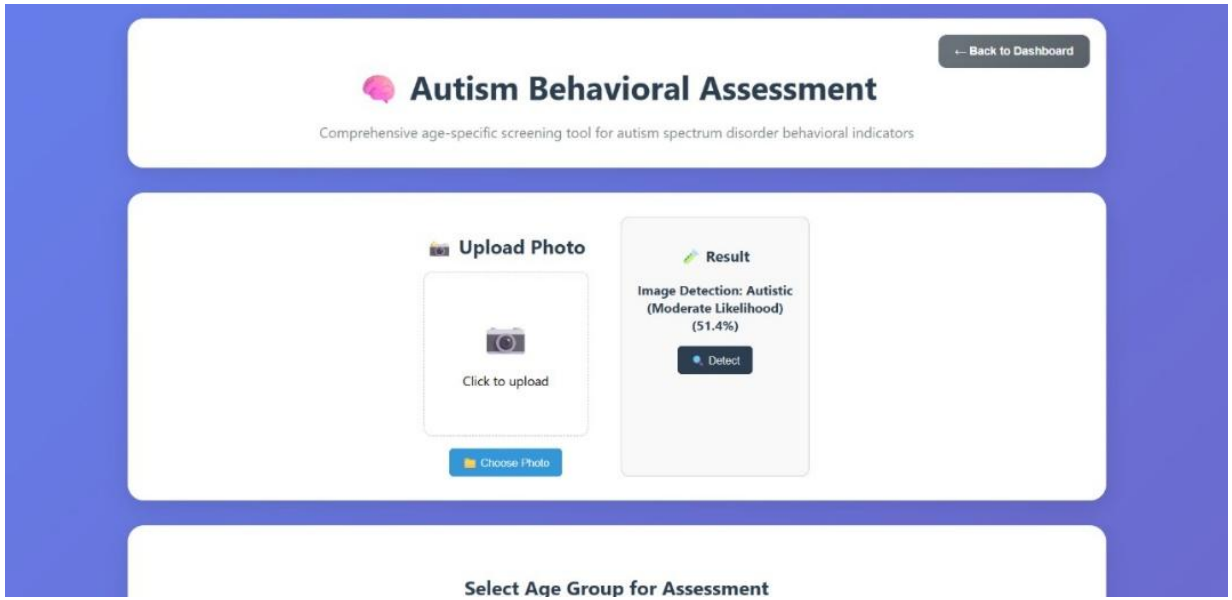
Implementation Status

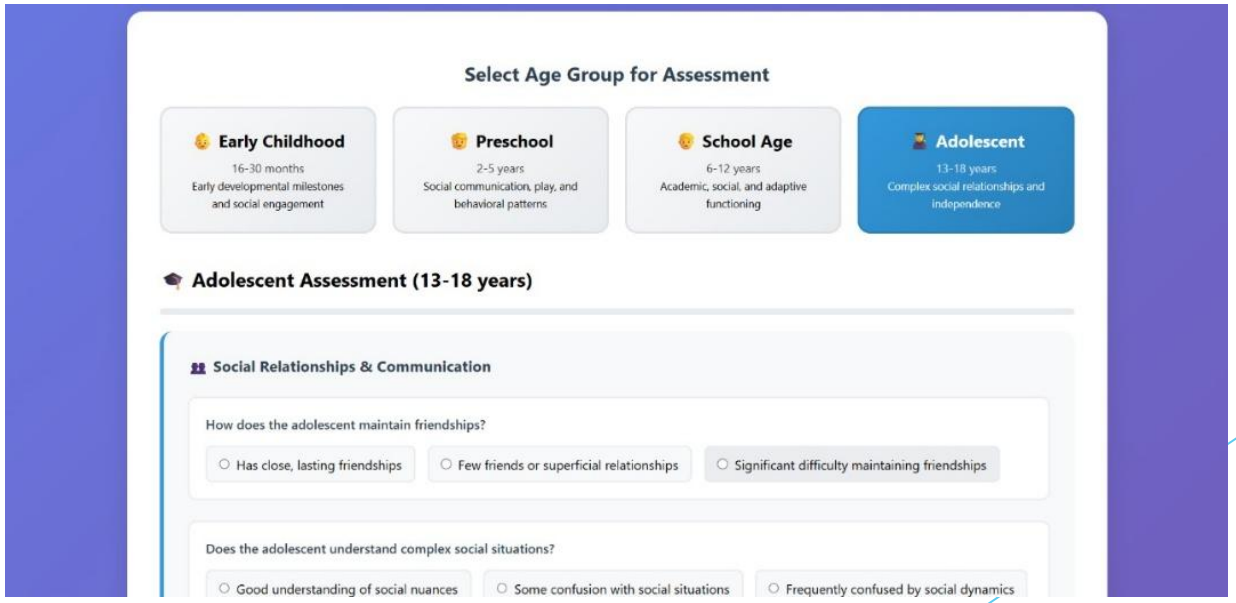




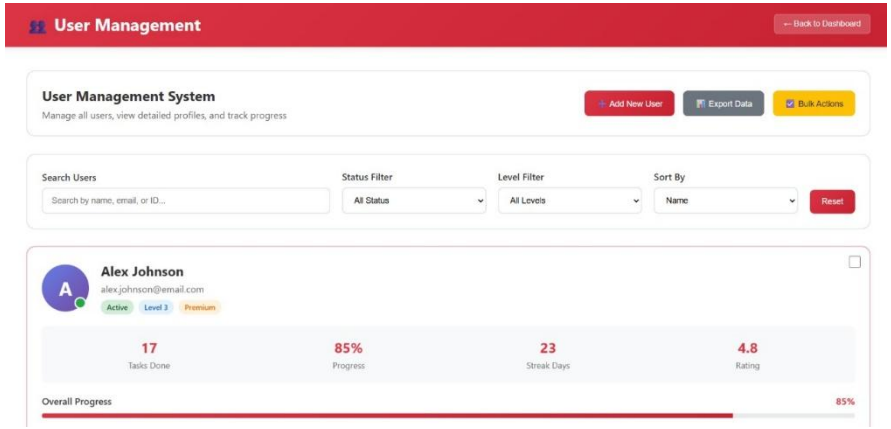


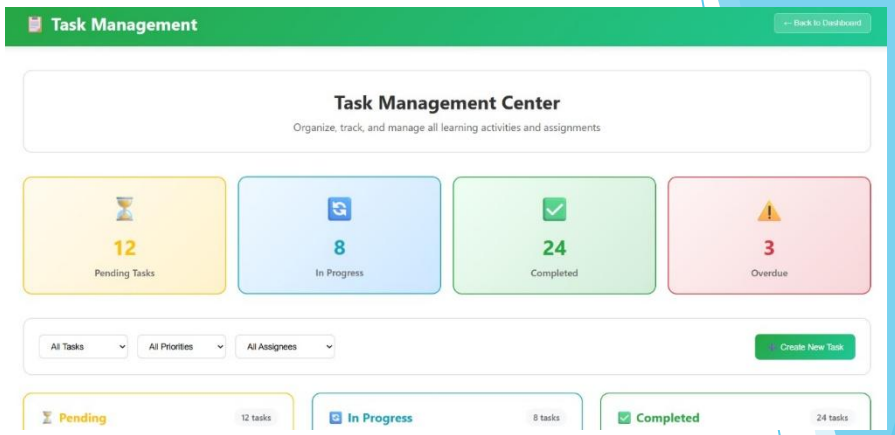
User Interface

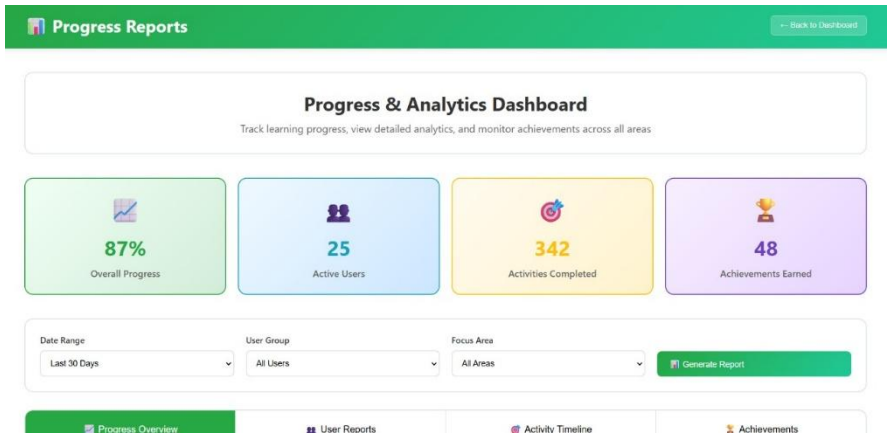


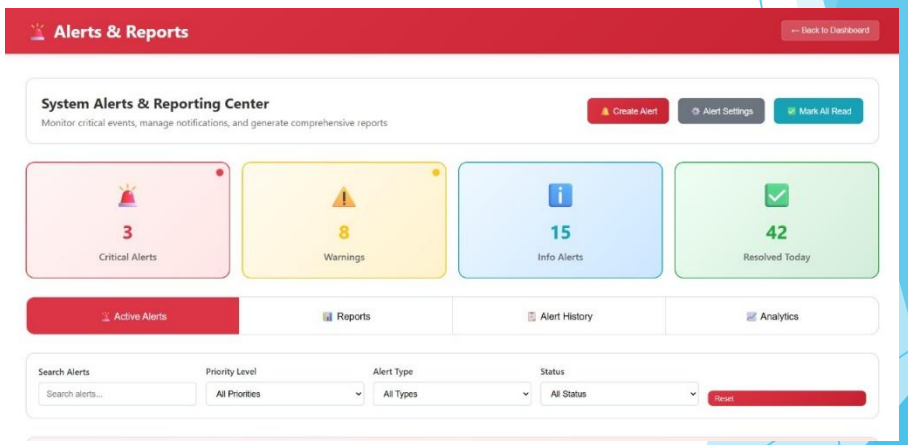


Admin Interface









References

[1] Koegel, L.K., Ponder, E., Bruzzese, T. *et al.* Using Artificial Intelligence to Improve Empathetic Statements in Autistic Adolescents and Adults: A Randomized Clinical Trial. *J Autism Dev Disord* (2025)

[2] Altın, Y., Boşnak, Ö. & Turhan, C. Examining Virtual Reality Interventions for Social Skills in Children with Autism Spectrum Disorder: A Systematic Review. J Autism Dev Disord (2025)

[3] R. S. Ali Fathima, R. Gokkul, M. Nirmalsahaj, S. Rohith and S. Rupesh, "AI-Powered Autism Detection, Awareness, and Social Integration: A Comprehensive Framework," 2025 International Conference on Computational Robotics, Testing and Engineering Evaluation (ICCRTEE), Virudhunagar, India, 2025.

[4] Fernandez-Lanvin, D., Gonzalez-Rodriguez, M., De-Andres, J. et al. Towards an automatic early screening system for autism spectrum disorder in toddlers based on eye-tracking.(2024)

References

[5] Parvej, Badhon, et al. "Computer Vision-based Interactive Autism Detection System using Deep Learning." 2024 IEEE International Conference on Computing, Applications and Systems (COMPAS). IEEE, 2024.

[6] Nagamani, Cheruku, et al. "Comparative exploration of machine learning models for enhanced autism detection." 2024 International Conference on Inventive Computation Technologies (ICICT). IEEE, 2024.

[7] Vakadkar, K., Purkayastha, D. & Krishnan, D. Detection of Autism Spectrum Disorder in Children Using Machine Learning Techniques. SN COMPUT. SCI. 2, 386 (2021).

Thank You...!!